

assignment _projectile

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:30pm

Quiz Instructions

2 attempts

There is a tutorial in the shared folder.

Make sure to go over the textbook (vector and projectile motion). Do the solved problems from the textbook. (hide solutions first)



Question 1 7.7 pts

While an object is in projectile motion (with upward being positive) with no air resistance

☐

the vertical component of its velocity remains constant and the vertical component of its acceleration is equal to $-g$.

☐

the horizontal component of its velocity remains constant and the vertical component of its acceleration is equal to $-g$.

☐

the vertical component of both its velocity and its acceleration remain constant.

☐

the horizontal component of its velocity remains constant and the horizontal component of its acceleration is equal to $-g$.

☐

the horizontal component of its velocity remains constant and the vertical component of its acceleration is equal to zero.



Question 2 7.7 pts

A hockey puck slides off the edge of a table with an initial velocity of 28.0 m/s. and experiences no air resistance. The height of the tabletop above the ground is 2.00 m. What is the angle below the horizontal of the velocity of the puck just before it hits the ground?

hint: off the table means horizontal initial velocity. $V_{ox}=28$ and $V_{oy}=0$

☐

31.8°

☐

12.8°

☐

72.6°



77.2°



12.6°



Question 3 7.7 pts

An object has a position given by $\vec{r} = [2.0 \text{ m} + (3.00 \text{ m/s})t]\hat{i} + [3.0 \text{ m} - (2.00 \text{ m/s}^2)t^2]\hat{j}$, where all quantities are in SI units. What is the magnitude of the acceleration of the object at time $t = 2.00 \text{ s}$?

0.00 m/s²0.522 m/s²2.00 m/s²1.00 m/s²4.00 m/s²

Question 4 7.7 pts

A catapult is tested by Roman legionnaires. They tabulate the results in a papyrus and 2000 years later the archaeological team reads (distances translated into modern units): Range = 0.2 km angle of launch = $\pi/4$ (45 degrees) landing height = launch height. What is the initial velocity of launch of the boulders if air resistance is negligible?



1.4 m/s



0.69 m/s



44 m/s



22 m/s



Question 5 7.7 pts

The components of vectors $\vec{\mathbf{B}}$ and $\vec{\mathbf{C}}$ are given as follows:

$$B_x = -9.2 \quad C_x = -4.5$$

$$B_y = -6.1 \quad C_y = 4.3$$

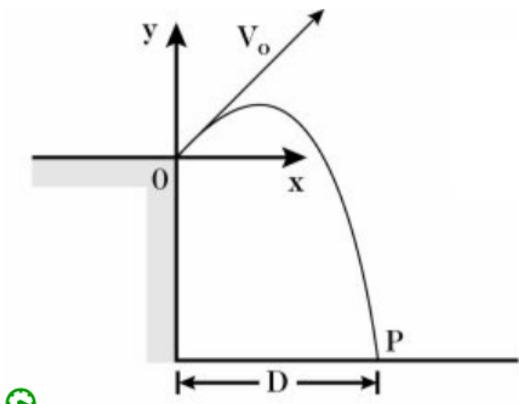
The angle (less than 180°) between vectors $\vec{\mathbf{B}}$ and $\vec{\mathbf{C}}$ is closest to

hint: use the dot product. $B_x C_x + B_y C_y = |\mathbf{B}| |\mathbf{C}| \cos(\text{angle})$
solve for angle.

- ☐ 84° .
- ☐ 10° .
- ☐ 77° .
- ☐ 170° .
- ☐ 103° .
- ☒

Question 6 7.7 pts

A projectile is fired at time $t = 0.0\text{s}$, from point 0 at the edge of a cliff, with initial velocity components of $V_{ox}=80\text{m/s}$ and $V_{oy}=200\text{m/s}$. The projectile rises, then falls into the sea at point P. The time of flight of the projectile is 50s



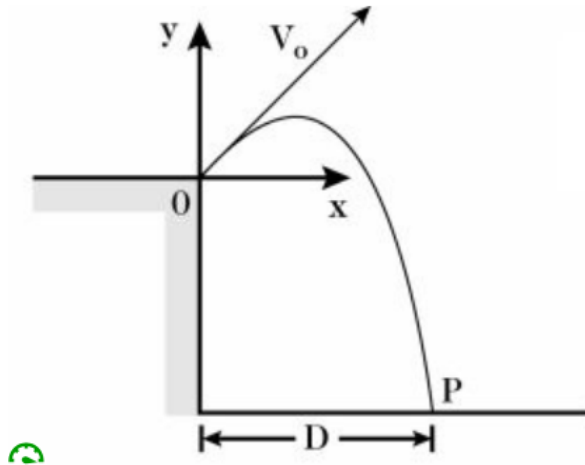
In Figure 3.2b, the x-coordinate of the projectile when its y-component of velocity equals 160m/s upward is closest to:

- ☐ 310 m
- ☐ 330 m
- ☐ 340 m
- ☐ 280 m
- ☐ 290 m



Question 7 7.7 pts

A projectile is fired at time $t = 0.0\text{s}$, from point 0 at the edge of a cliff, with initial velocity components of $V_{ox} = 30\text{m/s}$ and $V_{oy} = 100\text{m/s}$. The projectile rises, then falls into the sea at point P. The time of flight of the projectile is 25s



In Figure 3.2a, the magnitude of the velocity at time $t = 15.0$ is closest to:

- ☐ 47 m/s
- ☐ 247 m/s
- ☐ 49 m/s
- ☐ 56 m/s



249 m/s



Question 8 7.7 pts

A plane has an eastward heading at a speed of 156 m/s (relative to the air). A 20.0 m/s wind is blowing southward while the plane is flying. The velocity of the plane relative to the ground is

155 m/s at an angle 7.36° east of south.157 m/s at an angle 7.31° east of south.155 m/s at an angle 7.36° south of east.157 m/s at an angle 7.31° south of east.157 m/s at an angle 7.36° south of east.

Question 9 7.7 pts

A rock is thrown at a window that is located 18.0 m above the ground. The rock is thrown at an angle of 40.0° above horizontal. The rock is thrown from a height of 2.00 m above the ground with a speed of 30.0 m/s and experiences no appreciable air resistance. If the rock strikes the window on its upward trajectory, from what horizontal distance from the window was it released?



48.7 m



71.6 m



29.8 m



27.3 m



53.2 m



Question 10 7.7 pts

A boy throws a rock with an initial velocity of 2.15 m/s at 30.0° above the horizontal. If air resistance is negligible, how long does it take for the rock to reach the maximum height of its trajectory?



0.215 s



0.303 s



0.110 s



0.194 s



Question 11 7.7 pts

What is the maximum distance we can shoot a dart, from ground level, provided our toy dart gun gives a maximum initial velocity of 2.78 m/s and air resistance is negligible?



1.39 m



0.394 m



0.789 m



1.58 m



Question 12 7.7 pts

For general projectile motion, when the projectile is at the highest point of its trajectory



its acceleration is zero.



the horizontal component of its velocity is zero.



its velocity and acceleration are both zero.



the horizontal and vertical components of its velocity are zero.



its velocity is perpendicular to the acceleration.



Question 13 7.7 pts

Alice and Tom dive from an overhang into the lake below. Tom simply drops straight down from the edge, but Alice takes a running start and jumps with an initial horizontal velocity of 25 m/s. Neither person experiences any significant air resistance. Compare the time it takes each of them to reach the lake below.



Alice and Tom will reach the surface of the lake at the same time.



Tom reaches the surface of the lake first.



Alice reaches the surface of the lake first.

Not saved

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